What Drives Academic Growth in Chicago Public Schools

What drives academic progress in Chicago's public schools? This piece starts with two natural explanations, a school's ISBE designation and its location in the city, and finds that neither separates schools cleanly. The share of low-income students does. That result motivates the second half, which holds poverty fixed and asks whether school climate, chronic absenteeism, and enrollment change still relate to growth. All analysis uses 2023-24 data.

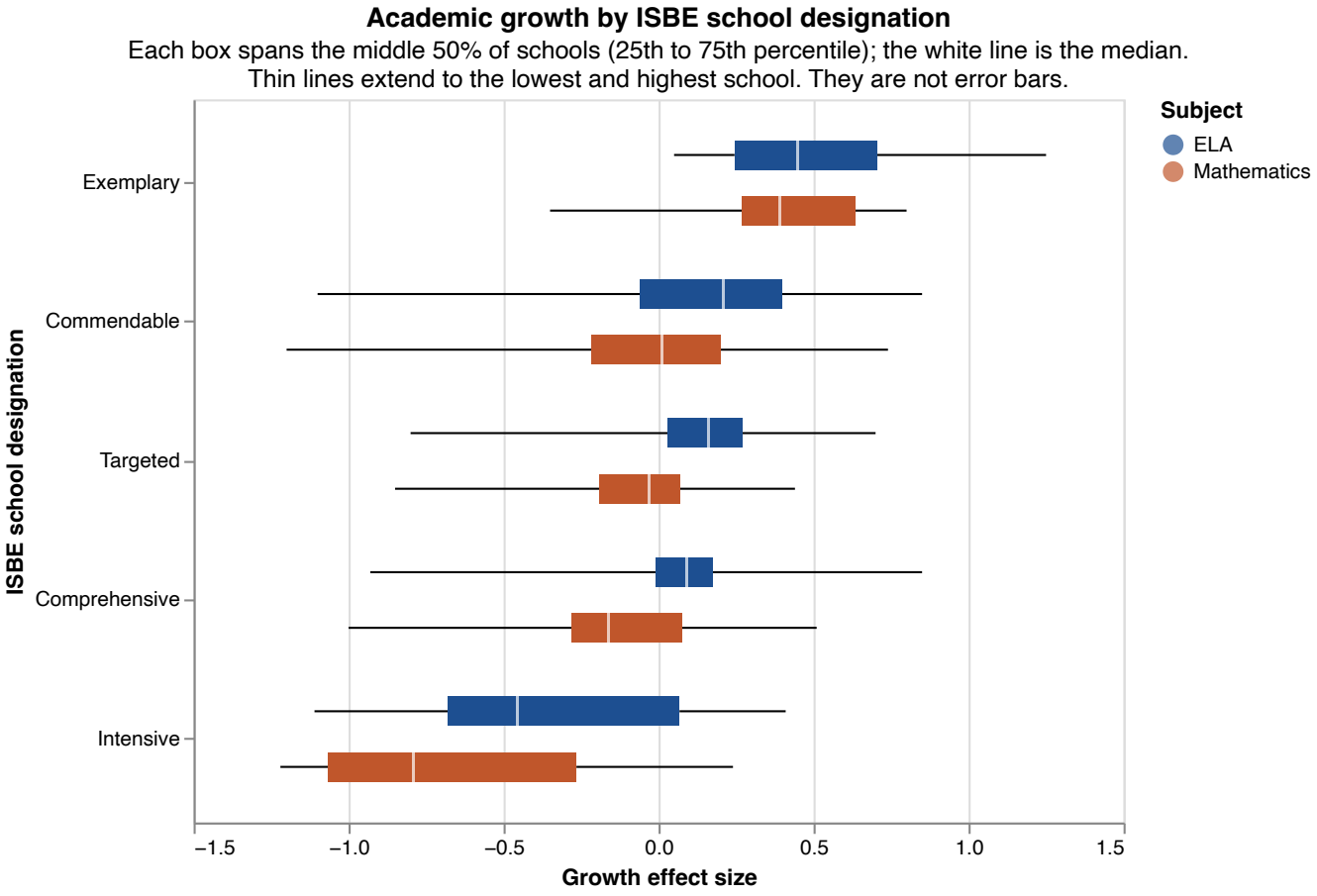
Background

Academic growth. The measure used throughout is the Academic Progress Indicator (API), CPS's student growth metric (CPS). It starts from each student's change in Normal Curve Equivalent (NCE) scores from one year to the next. An NCE is a standardized score from 0 to 100, where 50 is the state average. NCEs are computed from Illinois Assessment of Readiness scores in grades 3 to 8 and SAT/ACT scores in grades 9 to 11. Unlike percentiles, NCEs sit on an equal interval scale, so they can be averaged and compared across tests, grades, subjects, and years. Results are reported separately for English language arts (ELA) and mathematics.

Growth effect size. Each school gets one effect size per subject. It is the difference between the growth of the school's students and typical growth across the state for students who started at a similar level of achievement. A value of zero means students grew about as much as similar students statewide. Positive values mean more growth than similar students, and negative values mean less.

Low-income share. This is the percentage of a school's students classified as low-income, meaning they come from families with income within 185 percent of the federal poverty line (CPS). CPS adopted this term in place of "economically disadvantaged" and "free or reduced lunch eligible." In charts that compare groups, schools are ranked by this share and split into four roughly equal groups, from fewest to most low-income students.

Does a school's ISBE designation explain growth?



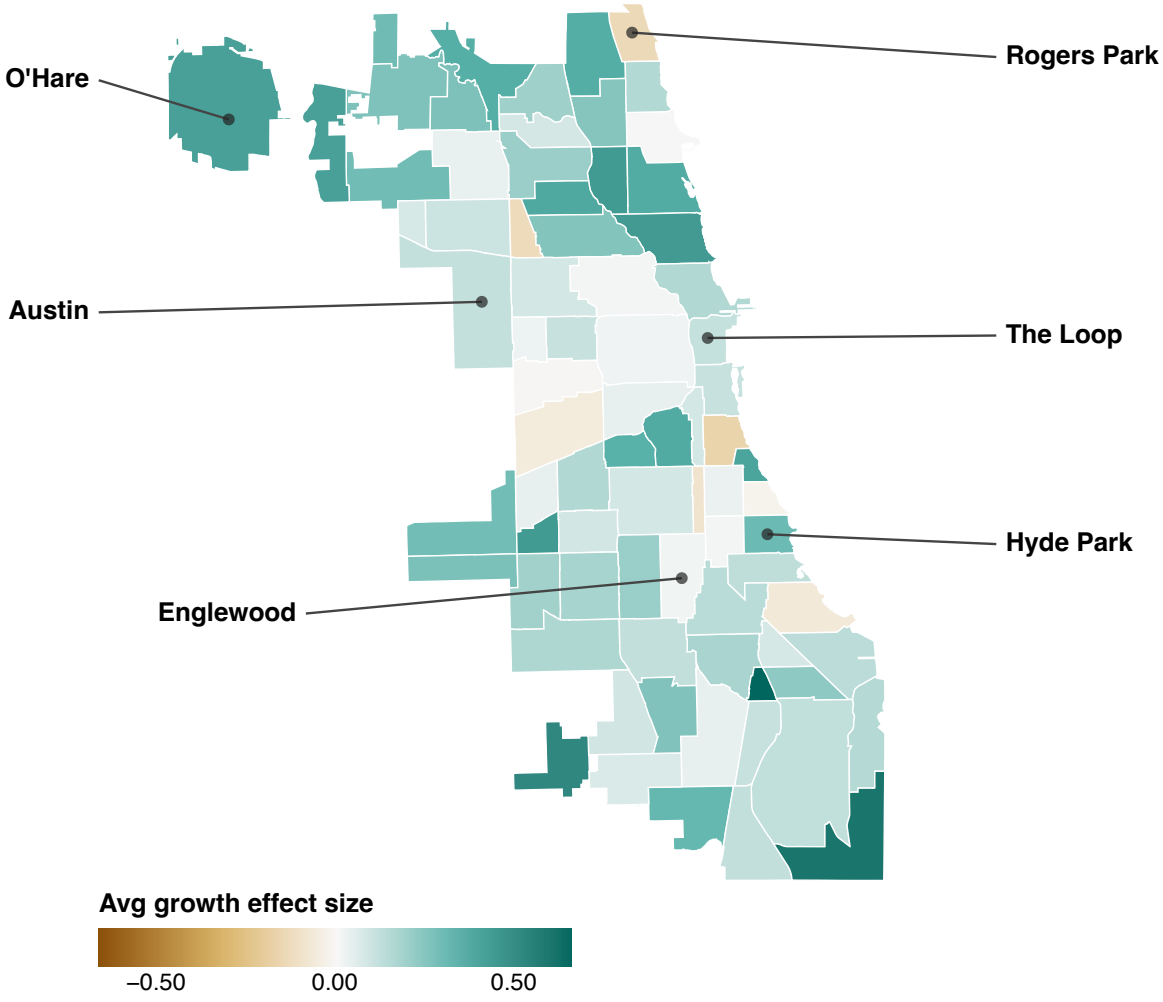
The Illinois State Board of Education (ISBE) gives each public school a summative designation every year under the federal Every Student Succeeds Act. The designation is based on the state's accountability system, specifically a multi-measures accountability index that compares school performance (ISBE). There are five tiers. *Exemplary* schools are in the top 10 percent statewide on that index and have no student group performing as poorly as the lowest 5 percent of schools. *Commendable* schools meet the same conditions but are not in the top 10 percent. *Targeted* schools have one or more student groups performing at that lowest-5-percent level. *Comprehensive* schools are in the lowest-performing 5 percent of eligible schools, and *Intensive* schools are those that stayed there after a full four-year improvement cycle. Roughly, Exemplary is the highest tier and Intensive the lowest.

Median growth does fall as designations get lower. In both subjects the median for Exemplary schools is positive, while for Intensive schools it is well below zero, and lowest in mathematics. But the pattern is far from clean. The boxes for the middle three tiers overlap heavily, and the thin lines show that Commendable, Targeted, and Comprehensive schools include both very high and very low growth. Only the two ends separate cleanly: Exemplary and Intensive schools rarely look alike. A school's designation therefore says little about how much its

students grew. Since designations are built from a performance index, some link with growth is to be expected, so a loose link is itself a reason to look for a better explanation.

Does location explain growth?

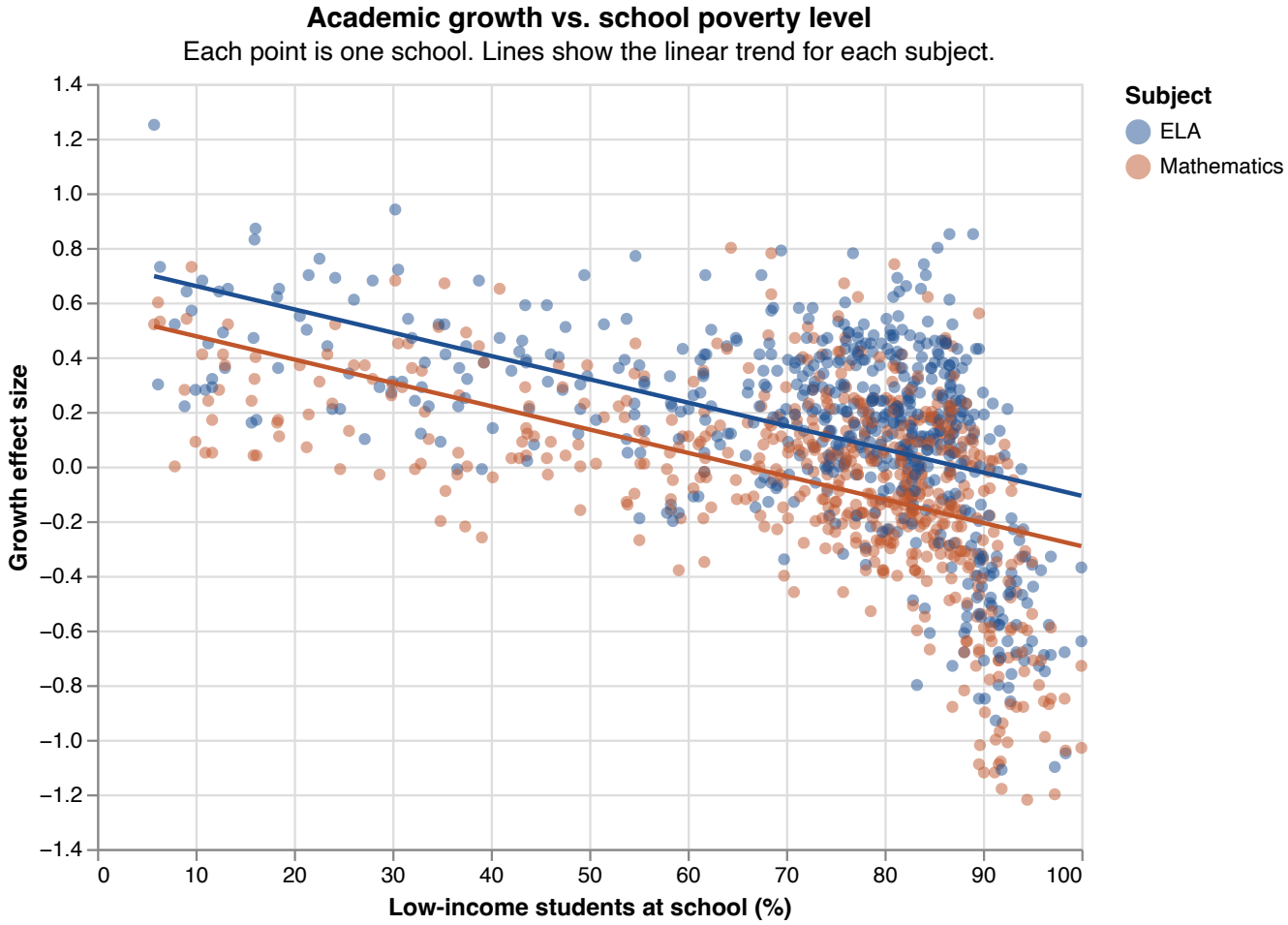
Average ELA academic growth by community area
Each area is colored by the average growth effect size of its schools.
Brown areas fall below zero and teal areas above. Labeled areas are for orientation.



A second natural explanation is geography. The map averages ELA growth effect size across the schools in each community area. Most community areas average above zero, with only a handful falling below it. The highest averages include Lincoln Park on the north side and Hegewisch and Burnside on the far south side, while the lowest include Rogers Park in the far north and Douglas and South Shore on the south lakefront. No single part of the city stands out as uniformly high or low, so location alone does not sort schools cleanly. These averages

should be read cautiously: the typical area contains only about six schools, and several of the highest values (Burnside, Hegewisch, West Elsdon) rest on one to three schools. The map shows ELA only, to keep it readable. The labeled areas are for orientation. Neither designation nor location sorts schools cleanly, so what does?

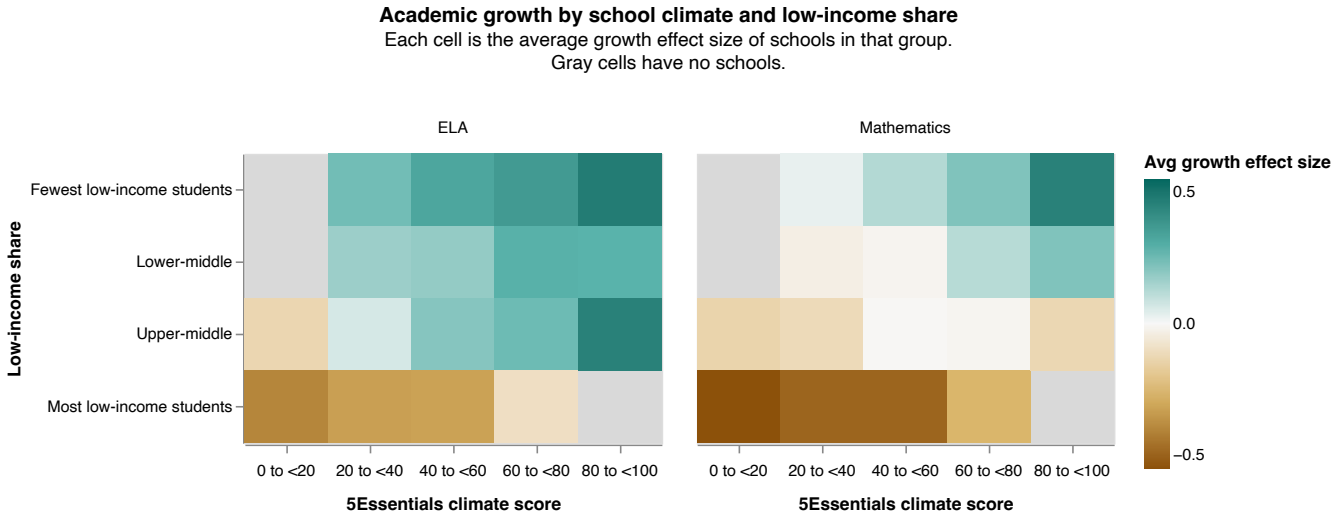
What does explain growth? School poverty



Growth does track the share of low-income students. Each point is a school (596 in total), and in both subjects the trend lines slope downward: the higher a school's low-income share, the lower its growth tends to be. The relationship is moderate, with a correlation of about -0.49 in ELA and -0.50 in mathematics. It is also uneven. Grouping schools into four equal groups by low-income share, the median ELA effect size is 0.32 for the group with the fewest low-income students and -0.35 for the group with the most. In mathematics the medians run from 0.13 to -0.44. The two middle groups stay close to zero or slightly above it, so most of the drop happens in the highest-poverty group, where the points on the chart fall away sharply above roughly 80 percent low-income.

School poverty therefore separates schools more consistently than either designation or location did. But it is an association, not an explanation. Poverty is tied to many other things that could shape growth. The rest of the analysis asks whether school climate, chronic absenteeism, and enrollment change still relate to growth when schools are compared within the same poverty group.

Does school climate matter once poverty is held fixed?



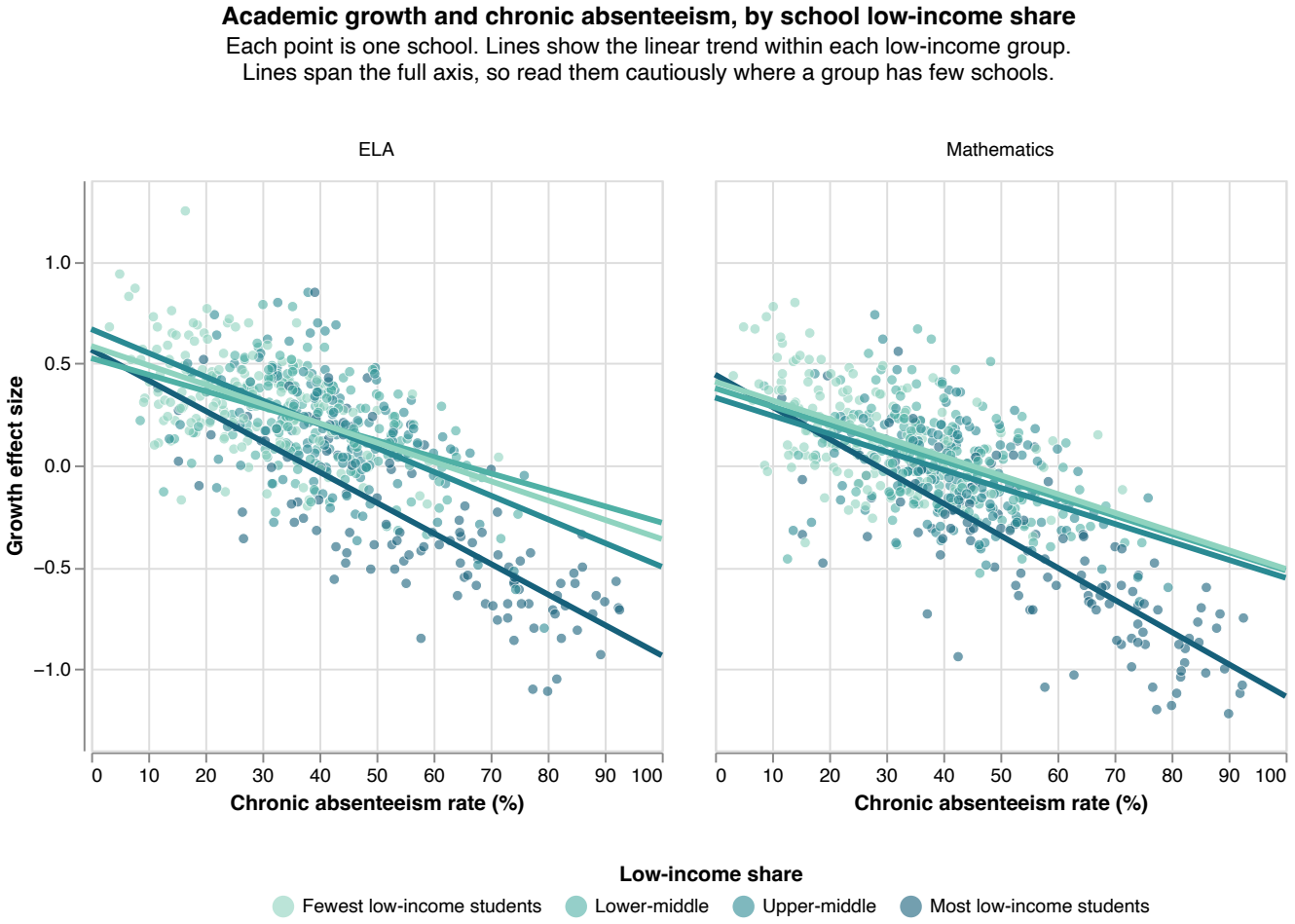
The 5Essentials survey, developed at the University of Chicago, measures a school's organizational conditions, including its culture and climate, across five factors: Effective Leaders, Collaborative Teachers, Involved Families, Supportive Environment, and Ambitious Instruction. It is given once a year to students and teachers, with an optional parent survey, and is used statewide in Illinois (UChicago Impact). This analysis averages a school's five 2024 essential scores into one climate score. The heatmap compares schools within the same low-income group, so each row holds poverty roughly constant and the columns show what climate adds. It covers 554 of the 596 schools; the rest have no climate score.

Within each low-income group, schools with higher climate scores tend to show more growth. In ELA, the group with the fewest low-income students rises from 0.24 in the 20-to-40 bin to 0.47 in the 80-to-100 bin, and the upper-middle group rises from 0.06 to 0.45. The shift is clearest where many schools sit: among the most low-income schools, those scoring 60 to 80 average -0.10 in ELA (24 schools) against -0.32 for those scoring 40 to 60 (73 schools), and -0.26 against -0.49 in mathematics. The pattern is not uniform, though. In mathematics the upper-middle group is roughly flat, between -0.12 and -0.01.

Climate does not erase the poverty gap. In every climate bin where both groups have schools, the group with the fewest low-income students outperforms the group with the most by about 0.5 in ELA,for example 0.32 against -0.32 in the 40-to-60 bin, which holds 330 of the 554 schools. Some cells rest on very few schools. The two ends

of the climate scale hold only 2 schools (0 to 20) and 14 schools (80 to 100), and several cells contain between one and four schools, so single cells should be read cautiously. Gray cells have no schools. Climate and poverty also overlap: none of the schools with the most low-income students scores above 80.

Does chronic absenteeism matter once poverty is held fixed?



Chronic absenteeism is measured as the number of chronically absent students divided by the school's enrollment, multiplied by 100. Students count as chronically absent as defined in Section 26-18 of the Illinois School Code, and medically homebound and hospitalized students are excluded (ISBE). The chart plots each school's rate against its growth (582 schools), with a separate trend line for each low-income group.

Higher absenteeism goes with lower growth, and the relationship holds within every low-income group. Across all schools the correlation is about -0.72 in ELA and -0.73 in mathematics, stronger than the link with low-income share alone. Within groups, the correlation ranges from about -0.43 to -0.53 in the three groups with fewer low-income students and reaches -0.77 in ELA and -0.80 in mathematics for the group with the most. The lines also

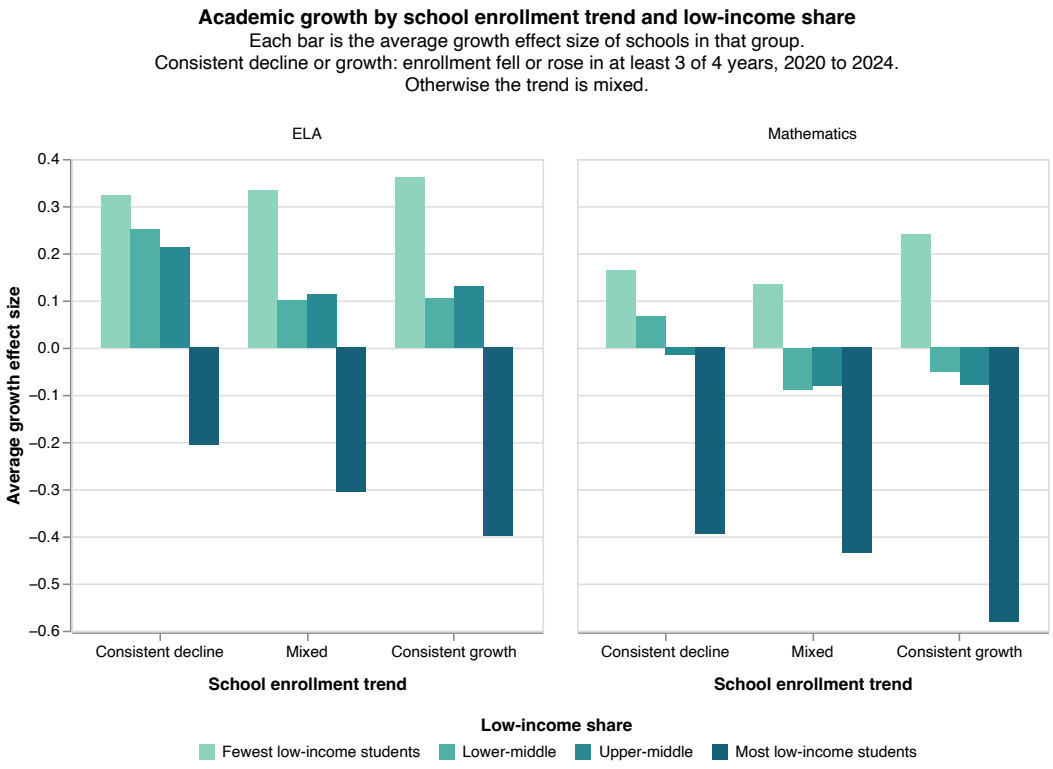
steepen with poverty. In ELA, each additional 10 points of absenteeism goes with about 0.09 less growth for the group with the fewest low-income students, 0.12 less for the upper-middle group, and 0.15 less for the group with the most. In mathematics the slopes are about 0.09, 0.09 and 0.16.

Three cautions apply. First, absenteeism and poverty overlap: the most low-income schools have the highest rates, up to 93 percent, while no school in the group with the fewest low-income students exceeds 71 percent. The trend lines run across the full axis for comparison, so the right-hand ends of the lower-poverty lines are extrapolated beyond the data. Second, the stronger overall correlation does not mean absenteeism matters more than poverty, because it likely reflects both. Third, absenteeism may be a symptom rather than a driver, since the same conditions that keep students out of school can also limit growth.

Why enrollment?

CPS runs more than 600 schools, and some of them are small and close to one another. The district faces repeated budget deficits, and enrollment decline has fed talk of consolidating schools. Funding has also changed. CPS ended student-based budgeting with its FY2025 budget, which had given each school a standard amount per enrolled student and favored larger schools. Schools now receive a fixed set of foundational staff, such as a principal, an assistant principal, a clerk, a counselor and core teachers, plus additional resources based on need (CPS, FY2025 budget fact sheet). Recent reporting says up to 120 assistant principal positions at small schools may be cut to help close the deficit (Block Club Chicago, May 2026). This analysis uses 2024 data, so none of those changes appear in it, but they are why enrollment is worth examining.

Does enrollment change matter once poverty is held fixed?



Schools are grouped by their enrollment trend from 2020 to 2024. A school counts as a consistent decline or consistent growth if enrollment fell or rose in at least three of the four yearly changes, and as mixed otherwise. Of the 581 schools with both growth and enrollment data, 335 declined, 168 were mixed and 78 grew. Each bar is the average growth of the schools in that trend and low-income group.

Schools with growing enrollment do not show more academic growth. In the three groups with more low-income students, schools with declining enrollment average equal or higher growth than growing schools in both subjects. Among the most low-income schools, declining schools average -0.21 in ELA and -0.40 in mathematics, while growing schools average -0.40 and -0.58. In the lower-middle group the ELA averages are 0.25 for declining schools and 0.10 for growing ones. Only among the schools with the fewest low-income students do growing schools come out slightly ahead, at 0.36 against 0.32 in ELA and 0.24 against 0.17 in mathematics. As a result, the poverty gap is widest in growing schools: in ELA the gap between the fewest and most low-income groups is about 0.76 for growing schools and 0.53 for declining ones.

Two cautions apply. The bars for growing schools rest on few schools, only 16 to 24 per group, so those averages are less certain than the ones for declining schools, which rest on 59 to 106. And this chart shows association only.

It does not say why growing schools do worse, and it cannot say whether consolidating schools would help, because it does not look at schools that were actually consolidated. A fuller answer would need to follow individual schools over time.

Conclusion

Neither of the two obvious explanations sorted schools cleanly. ISBE designation tracks growth, with the median falling from Exemplary to Intensive, but the middle tiers overlap heavily. Location shows no clear pattern across community areas. School poverty is the clearer divide: growth declines as the share of low-income students rises, and most of the drop is concentrated in the group with the most low-income students, whose median growth is below zero in both subjects.

The checks that hold poverty fixed point in the same direction. Within each low-income group, schools with higher climate scores tend to show more growth, and schools with higher chronic absenteeism show less. Absenteeism has the strongest link of anything examined here, and its penalty is steepest for the most low-income schools. Neither factor closes the poverty gap, though: schools with the fewest low-income students outperform those with the most at almost every climate level. Enrollment change does not show a simple benefit either. Among the three higher-poverty groups, schools with growing enrollment average equal or lower growth than schools with declining enrollment, so the poverty gap is widest where enrollment is rising.

These results are associations, not causes, and they have limits. Poverty, absenteeism and climate overlap, so the charts cannot say which one is doing the work. Growth is measured for a single year (2023-24), and the climate and enrollment measures come from other years. The enrollment chart does not study schools that were actually consolidated, so it cannot say whether consolidation would help. The most useful next step would be to follow individual schools over time, before and after changes in enrollment or funding, rather than comparing different schools in one year.

Data Sources

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